

CONEPRO · REQUIREMENTS

ConePro requirements

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DOCUMENT CONTROL

Document ID	CNP-REQ-001	Version	0.2
Status	Draft	Date	2026-09-25
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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/conepro		
Source	docs/03-requirements.md @ 69d559a		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

ConePro requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design (see CNP-PRB-001). One requirement, R6 (verticality warning), is **not met** by the current concept. Four more are at risk: R2 (depth accuracy), R3 (blow detection), R9 (test time, because the extraction method is open) and R11 (shock survival).

Reference test. A single test point driven to 850 mm in a medium-stiff soil averaging 15 mm per blow (about 57 blows), by one person, on level ground.

ID	Requirement	Target	Status at TRL 2	Verification (TRL 3 or later)
R1	Standard DCP geometry	8.0 kg hammer, 575 mm free drop, 16 mm rod, 20 mm cone with a 60 degree point, replaceable cone; tolerances as ASTM D6951	Met by design	Drawing check against the standard
R2	Depth measurement	Resolution 0.5 mm or finer; error within ± 1 mm over a 1,000 mm range compared with a steel rule	At risk: a low-cost draw-wire reel may reach about ± 2 mm (estimate)	Error budget calculation; later bench calibration
R3	Blow detection	Every blow counted and no false counts from handling, over a 200-blow sequence	At risk: sensor sits on the anvil in the impact shock	Design review; later bench test
R4	Per-blow record	Depth after each blow logged within 0.5 s of impact, with blow number and time	Met by design	Firmware sketch review
R5	Penetration range	850 mm or more per lower rod; 1,000 mm or more with one extension rod and automatic re-zero	850 mm met by the massing model (estimate); re-zero unverified	Model check; later field test
R6	Verticality	Warn the operator when the rod is more than 5 degrees off vertical	Not met: the concept has no tilt sensing on the rod	Design change needed (see CNP-PRC-001)
R7	Phone output	Live plot of depth against blows; DCP index per layer; indicative CBR from the ASTM D6951 correlations with the soil type selected; CSV export with time, location and operator notes	Met by design (app not written)	App specification review
R8	Works offline	No cell signal needed; logger stores 50 or more tests if the phone is absent or disconnects	Met by design	Firmware sketch review
R9	One-person test time	Reference test, including setup and rod extraction, in 12 min or less by one person	At risk: about 8 to 10 min estimated, but the extraction method is not chosen	Time estimate; later timed trials
R10	Portability	Total mass 16 kg or less; heaviest piece 10 kg or less; packed length 1.1 m or less	Met by estimate, thin margins: about 15.2 kg, heaviest piece about 9.8 kg, packed length about 1.0 m	Mass roll-up from the TRL 3 model
R11	Rugged electronics	Logger, sensors and connectors IP65; operate at 0 to 45 °C; anvil-mounted parts survive 10,000 blows	At risk: impact shock at the anvil estimated at hundreds to a few thousand g	Shock estimate; later drop-count test
R12	Battery life	8 h or more of continuous logging on user-replaceable non-lithium cells	Met by estimate: about 40 h on three AA alkaline cells	Power budget
R13	Cost	\$400 or less in parts for one complete instrument	Met by estimate: about \$296	Priced BOM (bom/ bom.csv)
R14	Honest results	App states the correlation used and that results are indicative and not for foundation design	Met by design	App specification review

■ Assumptions

- Soil behavior, blow rate and extraction time are estimates, not measurements. A practiced operator drops the hammer about once every 2.5 s.
- CBR correlations are those given in ASTM D6951 (from US Army Corps of Engineers work); other soils and regions may need local correlations.
- Mass estimates use steel at 7,850 kg/m³ and aluminum at 2,700 kg/m³ applied to the massing model.
- Battery life assumes an average electronics draw of about 50 mA and about 2,000 mAh usable from AA alkaline cells at that current.